

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

Let's break this into parts:

Part (a)

- **Quantity of heat required to heat 1 liter (1000g) of H₂O from 30°C to 60°C**

- Specific heat of water, $(c_{\text{H}_2\text{O}} = 1 \text{ cal/g}^\circ\text{C})$

- Mass of water, $(m = 1000 \text{ g})$

- Temperature change, $(\Delta T = 60^\circ\text{C} - 30^\circ\text{C} = 30^\circ\text{C})$

- Heat required, $(Q = m \cdot c_{\text{H}_2\text{O}} \cdot \Delta T = 1000 \cdot 1 \cdot 30 = 30000 \text{ cal})$

- Convert calories to kilocalories: $(30000 \text{ cal} \cdot \frac{1 \text{ Kcal}}{1000 \text{ cal}} = 30 \text{ Kcal})$

Part (b)

- **Quantity of heat required to heat 1 kg (1000 g) of aluminum from 30°C to 60°C**

- Specific heat of aluminum, $(c_{\text{Al}} = 0.215 \text{ cal/g}^\circ\text{C})$

- Mass of aluminum, $(m = 1000 \text{ g})$

- Temperature change, $(\Delta T = 60^\circ\text{C} - 30^\circ\text{C} = 30^\circ\text{C})$

- Heat required, $(Q = m \cdot c_{\text{Al}} \cdot \Delta T = 1000 \cdot 0.215 \cdot 30 = 6450 \text{ cal})$

- Convert calories to kilocalories: $(6450 \text{ cal} \cdot \frac{1 \text{ Kcal}}{1000 \text{ cal}} = 6.45 \text{ Kcal})$

Final Calculation

Combining both parts, the quantities of heat required are (30 Kcal) for the first part and (6.45 Kcal) for the second part, making the answer (G) .

<answer> G </answer>